Twin-DP3: View-Invariant 3D Diffusion Policy With Digital Twin

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***Abstract*—Learning visuomotor policies with imitation learning from 3D observations is a primary research direction in robotic manipulation, as 3D data inherently captures spatial features critical for action. While many existing methods rely on multi-view point cloud fusion, recent studies like 3D Diffusion Policy have shown that using a single-view observation setting with a lightweight point cloud encoder can achieve robust 3D policy learning. This design yields a simpler and more efficient pipeline. However, we observe that such a setting suffers from performance degradation under significant viewpoint changes. To overcome this limitation without requiring extensive multi-view data acquisition, we propose a digital twin-based real-time point cloud augmentation method named Twin-DP3. Our approach leverages a 6-DoF pose estimation algorithm to track the manipulated object and uses a robot model to generate robot component point clouds in real time. We validate the effectiveness of this method through comprehensive experiments in both simulation and real-world environments, demonstrating notable improvements in policy robustness under novel viewpoints.**

***Index Terms*—3D Visuomotor policy, viewpoint generalization, digital twin.**

I. Introduction

RECENT years have seen substantial breakthroughs in the application of imitation learning to robotic manipulation [1], [2], [3], [4]. Moreover, visuomotor policy learning from 3D observations is emerging as a key research focus in robotic manipulation due to the rich geometric information. Currently, many methods [3], [4] rely on globally fused point clouds from multi-camera systems. These global point clouds provide the most comprehensive observation information and are generally more effective in enhancing model performance. However, deploying multi-camera setups in real-world applications can be costly for demonstration collection and task execution. In contrast, 3D Diffusion Policy (DP3) [2] utilizes point clouds generated from a single-frame RGB-D input, achieving better performance than diffusion-based policies using RGB or RGB-D observations alone. This demonstrates that single-camera point cloud observations can be effective.

However, similar to policies that take RGB images as observations, such single-view point cloud policies suffer from limited viewpoint generalization. Although DP3 has demonstrated that using point cloud observations can maintain policy robustness under small viewpoint variations, its performance still degrades noticeably when the viewpoint shift becomes large. We argue that under point cloud observations, the generalization issue primarily arises from the fact that single-view point clouds represent only partial observations of the scene objects, and the discrepancy of learned 3D features between partial views from different viewpoints can be substantial.

To address the viewpoint generalization problem for single-view point cloud observations, we propose a novel approach that leverages a digital twin to infer global point cloud representations from partial single-view observations. Specifically, we begin by analyzing the point cloud observations used in the DP3 model and decompose the informative components for policy learning into two categories: the objects and the robot. For the objects, we first employ a 3D generative or reconstruction model to infer each object's complete shape from partial observations, obtaining a corresponding mesh representation. Then, we apply a robust 6-DoF pose tracking model to continuously estimate the object's position throughout the manipulation process. By combining the predicted mesh with the tracked pose, we reconstruct the object's point cloud at each time step. For the arm and gripper, we utilize the real-time joint state information to infer their geometry and sample the corresponding point cloud. Finally, by aggregating the reconstructed object and arm components, we construct a global digital twin point cloud, which is then used as the observation for the policy.

We validate through experiments that replacing observation point clouds with digital twin point clouds significantly improves the generalization of DP3 under novel viewpoints while maintaining competitive performance in the training view. To this end, the main contributions can be summarized as follows:

- We analyze and highlight the viewpoint generalization issue for partial observation in point cloud-based policies.
- We propose a digital twin-based real-time augmentation approach that generates complete 3D observations.
- We demonstrate through both simulation and real-world experiments that our augmentation method significantly improves success rates under novel viewpoints, validating its effectiveness in enhancing the viewpoint generalization of single-view point cloud-based policies.

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II. Related Work

A. Policy Learning With 3D Representations

Learning visuomotor policies from 3D observations is a central problem in robot learning. Prior work learns spatially robust policies from multi-view images [5], [6], RGB-D inputs [7], 3D point clouds [2], [8], [9], [10], [11], voxels [12], [13], and radiance fields [14], [15]. In parallel, many studies [3], [4], [16], [17] extract semantics from 2D foundation models and inject them into 3D representations to fuse geometry and semantics.

Among 3D modalities, point clouds are especially effective for complex tasks [18]. DP3 [2] and iDP3 [8] show that strong policies can be learned with a simple encoder on single-view point clouds. Recent 3D vision–language–action models also adopt a single third-person view with point-cloud input [19], [20]. We argue this setting will grow in importance as models and datasets scale due to its simplicity, scalability, and compatibility. Building on DP3, we identify a viewpoint invariance challenge that arises with single-view point clouds. We address it with a real-time digital twin augmentation that enriches observations with complete and temporally consistent 3D information.

B. Handling the Viewpoint Generalization Problem

Robustness to viewpoint variation is a core challenge for generalizing visuomotor policies trained from 2D observations. Mainstream methods adopt an augmentation paradigm that trains with multi-view data and tests with a single view [21], [22], [23], [24], [25]. Multi-view signals are often synthesized from single-view datasets using point cloud projection [23] or novel view synthesis [24], [25]. Other work seeks viewpoint-invariant scene representations. One approach distills a multi-view encoder into a single-view encoder to improve generalization [26]. ReViWo [27] pretrains a world model that separates view-invariant and view-dependent featires and adapts the invariant features for policy learning.

Although 3D representations in a unified frame are normally rotation and viewpoint invariant, single-view partial observations still cause large drops under strong viewpoint shifts. Multi-view point clouds, as in large-scale datasets such as DROID [28], can mitigate this issue but require substantially more data and higher collection and training costs. In contrast, our goal is to achieve viewpoint generalization with minimal reliance on extensive multi-view supervision.

C. Digital Twin in Robotics

Using digital twins to enhance real-world data and model training is a promising direction. Prior work builds real-to-sim pipelines that reconstruct real scenes in simulation to enable controlled, scalable training and evaluation [29], [30], [31], [32]. Other studies use digital twins for data augmentation. RoboTwin [33] combines 2D foundation models and large language models to synthesize diverse training data and a real-world-aligned benchmark for dual-arm manipulation. Scalable Real2Sim [34] automates real-to-sim by leveraging pick-and-place interactions to extract object geometry and physical properties. G3Flow [9] augments observations for 3D Diffusion Policy with semantic flow from the digital twin rather than augmenting data alone.

III. Approach

A. Preliminary: 3D Diffusion Policy

The imitation learning process is formulated as behavior cloning (BC) for visuomotor control. Given a set of expert demonstrations for a task $\mathcal{D} = \{(o^{(i)},\, a_0^{(i)})\}_{i=1}^{N}$, where $o^{(i)} \in \mathcal{O}$ is the observation data and $a_0^{(i)} \in \mathcal{A}$ is the corresponding action, the objective is to learn a policy $\pi : \mathcal{O} \to \mathcal{A}$ from the expert data.

Visuomotor policy learning via direct regression or policy-gradient methods often struggles with multi-modal action distributions, high-dimensional action spaces, and compounding errors during deployment. Diffusion policy [1] addresses the problems by iteratively denoising noisy actions to produce diverse, stable high-dimensional samples.

Diffusion Policy models a conditional distribution over actions using a Denoising Diffusion Probabilistic Model (DDPM) [35]. The denoising process performs K steps to denoise action a_t^0 from random noise a_t^k:

$$a_t^{k-1} = \alpha\big(a_t^k - \gamma\, \varepsilon_\theta(o_t, a_t^k, k)\big) + \mathcal{N}(0,\, \sigma^2 I) \tag{1}$$

where $\mathcal{N}(0,\, \sigma^2 I)$ is Gaussian noise added at each step and ε_θ is the noise prediction network. α, γ, and σ are step-dependent parameters of the noise scheduler.

To train the noise prediction network, a data sample a_t^0 is selected from the dataset. DDPM is then applied to this sample to obtain the corresponding noise ε^k at timestep k. The objective of training is to predict the noise that was added to the original data.

$$\mathcal{L} = \mathrm{MSE}\big(\varepsilon^k,\ \varepsilon_\theta(o_t,\, a_t^0 + \varepsilon^k,\, k)\big) \tag{2}$$

3D Diffusion Policy (DP3) [2] conditions the same diffusion policy framework on a 3D scene embedding. The observation is a learned 3D feature with robot states. This richer 3D conditioning yields stronger robustness, enabling DP3 to generalize with far fewer demonstrations.

B. Analysis of Viewpoint Generalization Problem for DP3

DP3 generalizes well to modest camera perturbations and object variations as described in its paper, but its robustness under much larger rotations remains untested. This motivates our study of DP3's performance under wide-range perspective shifts. Therefore, we propose a more challenging testing setting as shown in Fig. 1. In this setting, training data is collected from the robot's head camera, while seven novel view cameras are placed around the tabletop, as illustrated in the figure. These novel viewpoints are used to evaluate the model trained solely from the head camera's perspective. The simulation experiments are conducted in the RoboTwin [33] benchmark, which provides synchronized multi-view RGB-D observations and supports precise control of camera placement and robot interaction.

We use two tasks from the RoboTwin benchmark to evaluate the viewpoint generalization of DP3. The policy is trained using data from the training view only. We then evaluate the trained policy under both the training view and each of the sampled novel views, reporting the success rate in each case. The results are shown in Fig. 1. The results show a clear degradation in policy performance under varying viewpoints, with larger viewpoint shifts leading to more significant performance drops, as illustrated in the figure.

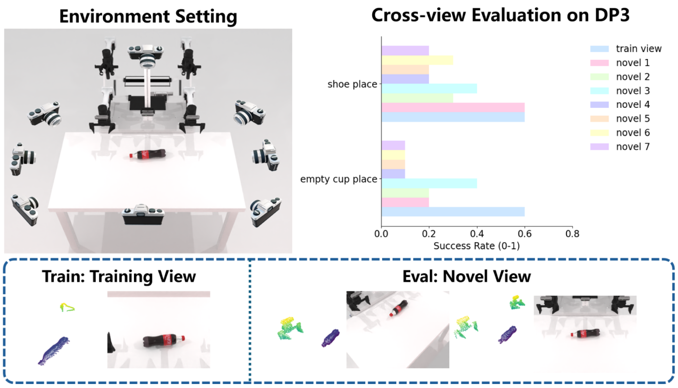


Fig. 1. Example of our experimental setting for evaluating viewpoint generalization. The policy is trained using a single-view point cloud observation, and tested under various novel views captured from multiple virtual cameras positioned around the scene. The evaluation of DP3 confirms its limited generalization under large viewpoint shifts.

We argue that the degradation in viewpoint generalization for 3D representations like point clouds primarily stems from the problem of partial observation. Under a single-view setting, both the target object and the robot are only partially observed, and the local observations from different viewpoints within the same scene can vary significantly. This discrepancy hinders the point cloud encoder from learning consistent geometric features. To address this issue, we propose replacing the partial point cloud observations with global point clouds constructed via a digital twin. By leveraging real-time 6-DoF pose estimation and inferring the robot's motion states, our method enables real-time acquisition of globally consistent 3D information. This enhances viewpoint generalization under partial observation settings.

C. Digital Twin Construction

We model the manipulation of rigid objects as a combination of the robot's motion and the motion of one or more objects in the scene. Under this assumption, the dynamic digital twin at each time step can be constructed by combining the point cloud of the robot (specifically the gripper) and the point clouds of the manipulated objects, both corresponding to their current states. The synthesis methods for these two types of digital twin, along with the DP3 pipeline conditioned on the twin point cloud, are illustrated in Fig. 2. We provide a detailed explanation in the following sections.

1) Object Digital Twin: The previous section identified partial observations of the object and the robot as the primary cause of DP3's limited viewpoint generalization. Accordingly, the objective of the object digital twin is to reconstruct a complete point cloud of the object and align it with the current 6-DoF pose estimated from observations.

The object digital twin generation process is inspired by prior digital-twin usage of G3Flow [9], which synthesizes object semantic flow using the digital twin. Specifically, we first reconstruct a 3D model of the object using a generative model, and then perform real-time 6-DoF pose estimation to obtain the object's bounding box under the current observation. Unlike G3Flow, which focuses on estimating the object's motion flow, our approach directly samples the 3D model at the predicted pose to generate a complete global point cloud of the object at each time step. Moreover, in contrast to G3Flow, which relies on a vision foundation model to extract semantic information and uses the resulting semantic flow as a condition for the diffusion policy, our approach is more straightforward. Experimental results show that our method achieves comparable performance with G3Flow under the training view, while significantly outperforming it under novel viewpoints.

2) Robot Digital Twin: We further posit that partial observations of the robot, particularly the gripper, constitute a significant factor limiting viewpoint generalization. Since the models of most robots and their grippers are readily available, and the joint states can be obtained in real time, we utilize forward kinematics to compute the robot's configuration at each time step. By sampling the mesh of the robot model under the current joint states, we obtain a point cloud representation of the robot in its real-time pose. Moreover, our ablation studies reveal that using only the digital twin of the gripper leads to the most noticeable improvement in policy performance.

3) Point Cloud Merging With Spatial Alignment: We assume that the training and novel view camera poses are pre-calibrated with respect to the robot base frame. Since the 6-DoF pose estimation of the object is performed in the camera coordinate frame, the point cloud sampled from the object's 3D model must undergo the following two spatial transformations. We denote $\mathbf{P}_o^{(i)}$ as the point cloud sampled from the model of $i-th$ object in the object frame, T_{oc} as the transformation of object pose predicted by 6-DoF pose estimator in the camera frame and T_{cw} as the camera to world transformation.

We first transform each object's point cloud to the camera coordinate frame:

$$\mathbf{P}_c^{(i)} = T_{oc}\mathbf{P}_o^{(i)} \tag{3}$$

For multi-object manipulation tasks, we fuse the point clouds of all manipulated objects in the camera coordinate frame and then transform the aggregated point cloud into the world coordinate frame, where '+' denotes point-set concatenation.

$$\mathbf{P}_c^{\text{objs}} = \mathbf{P}_c^{(1)} + \mathbf{P}_c^{(2)} + \cdots + \mathbf{P}_c^{(N)} \tag{4}$$

$$\mathbf{P}_w^{\text{objs}} = T_{cw}\mathbf{P}_c^{\text{objs}} \tag{5}$$

Finally, the robot's point cloud is integrated with the object point clouds in the world coordinate frame to form the final digital twin-augmented point cloud.

$$\mathbf{P}_w^{\text{twin}} = \mathbf{P}_w^{\text{objs}} + \mathbf{P}_w^{\text{robot}} \tag{6}$$

4) Deployment of Digital Twin in DP3: As illustrated in Fig. 2, during both training and evaluation, we do not directly use the raw point cloud generated from the observation. Instead, we infer the twin-augmented point cloud by the procedure described in the previous section. This enhanced representation is then fed into the subsequent modules of the 3D Diffusion Policy. Experimental results show that using the twin-augmented point cloud not only improves performance under the training view but also leads to a more substantial performance gain under novel views, achieving success rates that closely approach training view evaluation results.

IV. Experiments

We perform extensive experiments on the proposed digital twin augmentation paradigm in both simulation and real world.

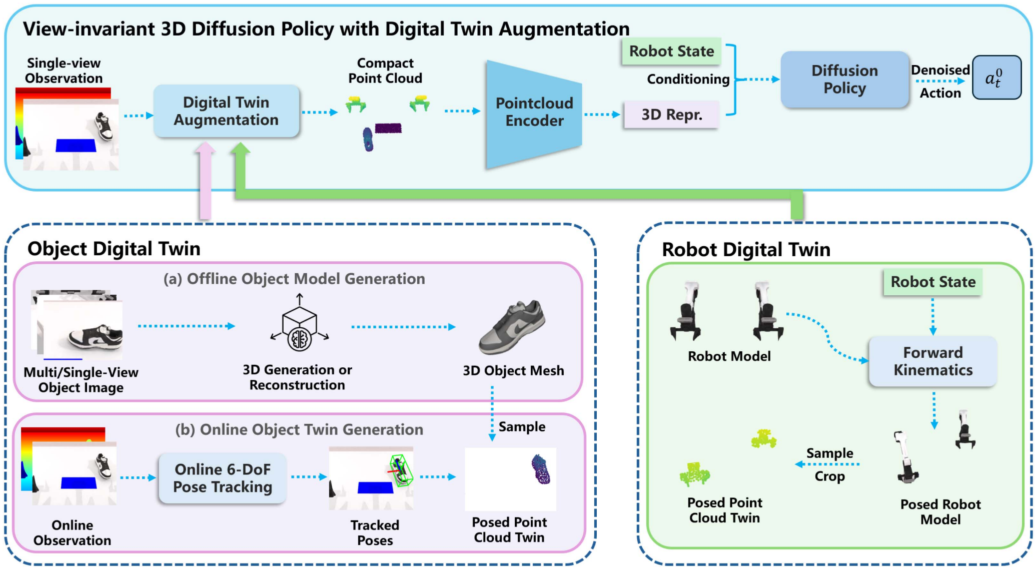


Fig. 2. **Overview of Twin-DP3. Above:** During both training and inference, the 3D visual observation is reconstructed using an online digital twin point cloud. This point cloud is encoded by a point cloud encoder into a compact 3D representation, which together with the robot state is used as the condition for the diffusion sampling process in the Diffusion Policy. **Below:** The digital twin generation consists of two modules. For the objects, we first generate 3D models for each object. These meshes are then used for online 6-DoF pose tracking and point cloud sampling, resulting in a posed object point cloud twin. For the robot, we apply forward kinematics to the robot model with the current robot state to generate the robot model in its present configuration. A corresponding point cloud is then sampled from this model. The digital twin point clouds of both the object and the robot are fused to form the observation input for the policy.

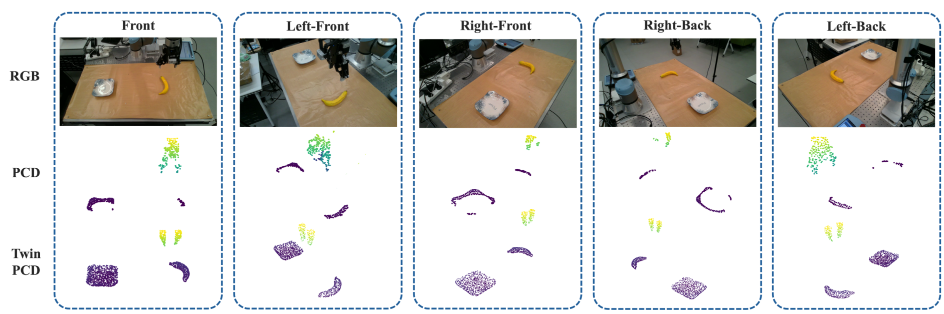


Fig. 3. **Real-world Experimental Setting.** First row: RGB images showing both training and novel view observations in real-world experiments. Second row: Observation point clouds reconstructed from RGB-D images. Third row: Digital twin point clouds generated using our proposed method.

We address the following questions: (1) Can our real-time digital twin effectively mitigate the viewpoint generalization issue caused by partial observations, and improve policy success rates across arbitrary viewpoints? (2) For policies conditioned on 3D observations, which parts of the robot should be preserved in the observation to best support task representation learning? (3) If using multi-view data for training, will the original DP3 achieve the same level of results as our method? (4) Analysis on the approach's object generalizability, scalability and latency.

Fig. 4. **Simulation Experiment Tasks**. The tasks include three single-arm and two dual-arm tasks.

A. Experimental Setup

In simulation, we evaluate our approach on five distinct tasks in the RoboTwin [33] benchmark with both single-arm and dual-arm tasks. We follow the benchmark's baseline to train the policy with 100 episodes and 3000 epochs in training view. Evaluation under training and novel view is conducted with 100 test episodes. We use the head camera as the training view and sample 7 novel viewpoints for evaluation around the workspace (as shown in Fig. 1). The tasks are depicted as: (1) **empty cup place**: place the cup on the mat; (2) **shoe place**: put a randomly placed shoe on a mat with toe facing left; (3) **bottle adjust**: grasp bottles based on opening orientation and put upright; (4) **diverse bottles pick**: position two bottles of varying sizes and shapes; (5) **shoes place**: position two shoes with synchronized bi-manual control. The task environments are shown in Fig. 4.

We evaluate our method in two real-world tasks including **banana and mug pick-and-place** with a UR5e robot. We set 5 camera viewpoints using an Intel RealSense L515 camera as shown in Fig. 3. The front view is set as the training view. For each task, we collect 40 demonstrations for training and evaluate for 10 episodes per view.

TABLE I
SIMULATION EXPERIMENT RESULTS (SUCCESS RATES IN **TRAIN/NOVEL** VIEWS)

Task	empty cup place	shoe place	bottle adjust	diverse bottles pick	shoes place	Train View Avg.	Novel View Avg.
DP3[2]	0.62/0.22	0.57/0.27	0.72/0.27	0.49/0.05	0.08/0	0.496(-0.124)	0.162(-0.408)
G3Flow[9]	0.75/0.19	0.75/0.39	0.81/0.66	0.62/0.14	0.2/0.06	0.626(-0.002)	0.288(-0.282)
Twin-DP3	0.75/0.56	0.77/0.71	0.84/0.87	0.66/0.63	0.12/0.08	**0.628**	**0.57**

For a fair comparison, we use the same demonstration data and only replace the observed point cloud with the digital twin-augmented point cloud for training. We use FoundationPose [36] as our 6-DoF object pose predictor.

B. Simulation Experiments

The experimental results in the simulation environment are shown in Table I. As observed across the selected tasks, DP3 exhibits a clear drop in success rate under novel views compared to the training view. This confirms the viewpoint generalization issue inherent in single-view 3D policy learning. G3Flow's semantic flow enhancement leads to a notable performance gain over DP3 under the training view. Our results further show that G3Flow also achieves measurable improvements in success rate under novel views. However, the performance gap between the training view and novel views remains substantial. In contrast, our method enables the success rate under novel views to closely match that of the training view. Specifically, the average success rate under novel viewpoints is only 5.8% lower than that under the training view. This represents a substantial improvement over both baselines, demonstrating the effectiveness of our approach. Moreover, our method achieves performance under the training view that is comparable to G3Flow, without any semantic inputs, highlighting its simplicity and effectiveness.

We attribute the advantage of our method to two key factors: (1) the preservation of complete object geometry, which ensures consistent spatial information across views, and (2) the removal of noise and policy-irrelevant parts from raw observations. Together, these lead to point clouds that are both cleaner and more informative for policy learning.

C. Real-World Experiments

We first visualize the comparison between the point clouds from observations and from the digital twin in Fig. 3. This reveals that RGB-D point clouds captured from different viewpoints retain different portions of the object and the robot arm and gripper, which leads to variation in the observed geometry. This variation persists, even when all point clouds are transformed into the world coordinate frame. In contrast, our method can reconstruct a nearly consistent scene point cloud from any viewpoint. Once aligned in the world frame, these point clouds provide the policy with a coherent understanding of the scene, enabling it to leverage complete observations to improve performance and enhance viewpoint invariance. In Table II, we report the success rates of the model trained under the front view when evaluated from the camera viewpoints shown in Fig. 3. Consistent with the results observed in the simulation environment, DP3 exhibits a noticeable drop in success rate when facing large viewpoint shifts. In some views, the policy succeeds in only 0 to 2 out of 10 trials under certain viewpoints. In contrast, the twin-augmented version of DP3 achieves success rates under novel viewpoints that are largely comparable to those under the training view. Across both tasks, the average success rate under novel viewpoints improves by a factor of 1–2× compared to the original DP3. These results demonstrate that our method effectively enhances the robustness of 3D Diffusion Policy in real-world tasks.

TABLE II
REAL-WORLD EXPERIMENT RESULTS (SUCCESS RATES)

Viewpoint	**Twin-DP3**		**DP3**	
	banana	mug	banana	mug
train view	8/10	9/10	6/10	9/10
left_front	7/10	7/10	3/10	5/10
right_front	7/10	9/10	4/10	1/10
left_back	8/10	7/10	4/10	0/10
right_back	8/10	6/10	2/10	4/10
Novel View avg.	**0.75**(+0.425)	**0.725**(+0.475)	0.325	0.25

We further perform a qualitative analysis of the robot's behavior in these test-time episodes. The examples are shown in Fig. 5. We observe various types of failure, including inaccurate positioning, gripper open/close errors, incorrect object placement, and even oscillatory movements of the robotic arm during execution. These issues highlight the limitations of relying solely on single-view RGB-D observations. In contrast, our digital twin-augmented policy maintains stable behavior across novel viewpoints and achieves success rates that are comparable to those under the training viewpoint, demonstrating improved viewpoint robustness.

D. Ablation Study

1) Novel View Ablation: We evaluate our method and the baselines at each of the sampled viewpoints illustrated in Fig. 1, and report the success rate for every individual viewpoint. The results are presented in Fig. 6. The results show that our method maintains stable success rates across the 360-degree novel viewpoints sampled around the workspace. In contrast, both baselines exhibit a significant drop in performance as the viewpoint deviates from the training view. This demonstrates that our augmentation provides strong 360-degree viewpoint robustness.

2) Part Ablation: To further investigate which components of the point cloud observation contribute most to policy learning and viewpoint generalization, we conduct an ablation study based on the content of the observed point cloud. We choose two tasks from RoboTwin for this study and the results are shown in Table III.

The experimental results show that using full point clouds from the digital twin greatly improves success rates under novel viewpoints. This holds for both the object and the embodiment. Under the training viewpoint, the digital twin also brings moderate improvements. In contrast, replacing any part of the digital

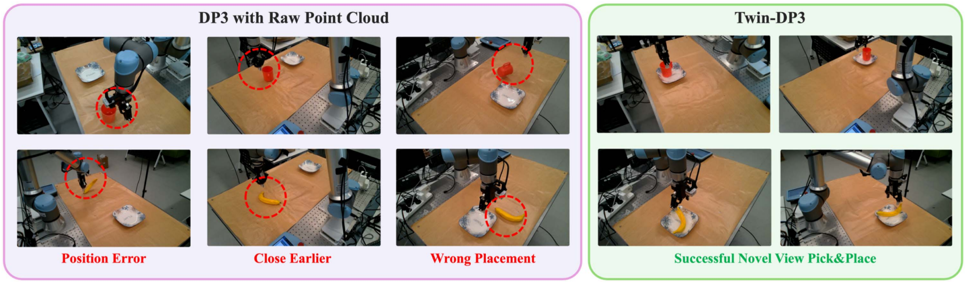


Fig. 5. **Example of real-world experiment results.** When the viewpoint is changed, DP3 exhibits various execution failures such as position errors, gripper open/close errors, incorrect object placement, and even oscillatory movements of the robotic arm. In contrast, when using the digital twin as the observation input, the policy achieves success rates under novel viewpoints that are close to those under the training view.

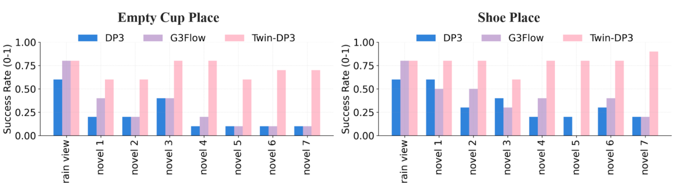


Fig. 6. **Ablation across Novel Viewpoints in Simulation.** Compared to the baselines, our digital twin approach achieves more stable success rates across all sampled novel viewpoints, demonstrating its effectiveness in improving viewpoint robustness.

Fig. 7. **Novel-view training comparison.** Even with additional training, DP3 remains below Twin-DP3, confirming the efficiency and viewpoint robustness afforded by the twin point cloud observation.

TABLE III
ABLATION OF POINT CLOUD COMPONENTS (SUCCESS RATES)

Task	**shoe_place**		**diverse_bottles_pick**	
Component/view	train	novel	train	novel
twin obj+gripper	**0.77**	**0.71**	**0.66**	**0.63**
twin obj+arm	0.60	0.58	0.52	0.48
twin obj	0.67	0.56	0.58	0.5
twin obj+obs arm	0.60	0.28	0.51	0.19
obs obj+twin arm	0.51	0.11	0.52	0.11
full obs	0.57	0.27	0.49	0.05

TABLE IV
TRACKING RESULTS OF OBJECT MODELS(AUC OF ADD-S(↑))

Used Model	**obj1**	**obj2**	**obj3**	**obj4**	**obj5**
Novel Model	0.9624	0.9598	0.9622	0.9572	0.9564
Training Model	0.8551	0.8431	0.9424	0.7206	0.5155

twin with the raw observation point cloud causes a clear drop in performance.

For the digital twin setting, we find that using the combined point cloud of the object and the gripper leads to higher success rates compared to using only the object or including the entire robotic arm. We consider this finding both reasonable and insightful, as it highlights the importance of preserving task-relevant geometry while avoiding unnecessary information that may hinder policy learning.

3) Multi-View Training Ablation: This section evaluates whether training the original DP3 directly with novel-view data can achieve the same success rate under novel views as our method. We select two tasks and train the policy with 100–800 demonstrations collected from novel viewpoints. After training, we evaluate the policy under the same novel viewpoints and report the success rates, as shown in Fig. 7.

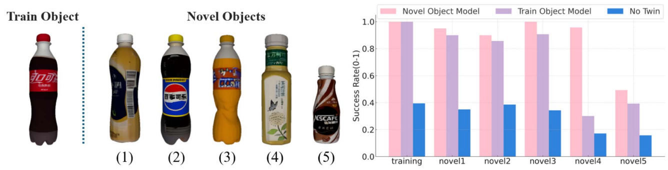
From the results, we observe that even when training directly with multi-view data and increasing the number of demonstrations to eight times that of the original setting, the final success rate under novel views still falls short of that achieved by our method. This gap may stem from the inability of a simple point cloud encoder to learn viewpoint-invariant representations from point clouds captured at different viewpoints. These findings further validate the positive effect of digital twin-generated global observations in facilitating representation learning for point cloud encoders.

V. Discussions and Limitations

A. Object Generalizability

Cross-object generalization is essential for deploying imitation-learning policies in real-world applications. Within our pipeline, both the tracking model and the policy model face generalization challenges. Zero-shot object model generation enables tracking and manipulation on novel objects, but per-object reconstruction remains costly. Category-level or model-free tracking can reduce this cost but may introduce inaccuracies in tracking and digital twin point cloud and degrade the policy. To study these trade-offs, we evaluate RoboTwin's bottle-adjust task. The policy setup and results are shown in Fig. 8. Training uses a single object and testing uses novel objects under three digital-twin conditions: reusing the training model, using each test object's model, and no digital twin. Tracking performance is reported in Table IV as the AUC of the ADD-S metric [37].

The results show that an object-specific digital twin yields the strongest policy and tracking generalization. Policy performance degrades on object (5) due to a large size change, which exposes



(a) Train and Test Object Models (b) Policy Results (Average in All Views)

Fig. 8. **Object Generalizability: Setup and Results.** We use a single object model to collect demonstrations and evaluate the policy on unseen objects. Evaluation covers three settings: a digital twin built from the novel object's model, a digital twin that reuses the training object's model, and no digital twin. Using a single object model could migrate to texture changes and moderate shape variation. Performance drops under large shape or size shifts due to mismatched contact geometry and reduced tracking reliability.

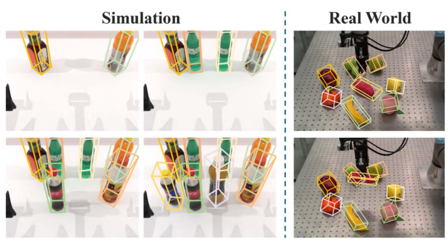


Fig. 9. **Example of Scalability Experiment. Left:** we test tracking latency, accuracy and policy success rate with an increasing number of objects with occlusion in simulation. **Right:** we also conduct qualitative experiments in the real world to assess the feasibility of multi-object tracking.

TABLE V
LATENCY, TRACKING AND POLICY RESULTS ON CLUTTERED SCENES

Num of Objs		2	4	6	8
Latency (ms)		83.41	89.21	93.78	97.23
Tracking	ALL	0.9213	0.8989	0.9096	0.9163
	Moving	0.9213	0.8603	0.8296	0.8355
Policy	Twin	0.95/0.95	0.95/0.95	0.9/0.9	0.85/0.85
	Non-Twin	0.9/0.5	0.85/0	0/0	0/0

limited policy robustness to large scale differences. The policy reusing the training object's twin remains stable on the first three novel objects where texture varies and shape changes are small, despite the moderate drop in tracking accuracy. It degrades on object (4), whose geometry shifts from a cylinder to a prism. The shape mismatch between the twin and the target harms both tracking and policy.

In summary, using general object models shows potential to replace novel objects in our pipeline, but robust deployment requires finer categorization by shape and size. Improving the policy model's generalization to objects within the same category under larger shape changes and reducing the reliance of tracking models on object-specific reconstructions as a model-free pipeline remain important future research directions.

B. Scalability

In this section we assess the scalability of our twin-augmented pipeline. We extend the diverse bottles pick task to cluttered multi-object scenes with simplified object pose and shape settings, as shown in Fig. 9. We report per-step latency, tracking accuracy, and policy success rate in Table V. Tracking is evaluated by the AUC of ADD-S. Policy is evaluated by success rates on training/novel views.

TABLE VI
INFERENCE TIME ANALYSIS

Process	time
Get Obs	60 ± 10 ms
Generate PCD from Obs	20 ± 2 ms
PCD Sampling(Grid)	10 ± 1 ms
Model Inference	40 ± 2 ms
Generate Twin	55 ± 10 ms

The results show that tracking and policy performance remain robust as the number of objects grows, though pronounced occlusions reduce accuracy. The digital twin provides an even larger advantage under occlusion. With occlusions on both training and novel views, DP3 fails to complete the task, while the twin-augmented policy maintains a solid success rate, demonstrating stronger resilience to more severe viewpoint-induced occlusions in cluttered scenes. Latency increases slightly with FoundationPose's parallel multi-object tracking, but the growth is sublinear and manageable. A practical optimization is to detect motion and track only moving objects while reusing prior poses for static ones, which limits the number of objects tracked in parallel.

C. Time Analysis

To evaluate the impact of incorporating digital twin inference on the runtime performance of DP3, we measure the execution time of several key steps in the real-time inference pipeline. The measurements are conducted on a workstation with an Intel i9-12900 CPU and an NVIDIA RTX 4090 GPU. The results are summarized in Table VI.

Camera data acquisition (Get Obs) runs as a separate thread in parallel with the model thread. The non-twin policy runs at about 70 ms per step with point cloud generation from RGB-D images, filtering and sampling. Twin generation runs in approximately 55 ms per frame. Model initialization and mesh-based point cloud sampling are performed offline before each episode, so no point cloud construction is needed at test time. The total per-step inference time remains about 95 ms. The added cost over raw observation DP3 with grid sampling is modest and still supports inference at about 10 Hz. Latency can be further improved by balancing model performance and speed, for example by decreasing the sampling and refinement iterations in FoundationPose or using a more lightweight diffusion U-Net for the diffusion policy.

D. Limitations

Our approach has several limitations. First, it assumes calibrated cameras in training and novel views. Second, it mainly relies on instance-level (or category level in similar shape and size) object models and tracking, so it does not yet generalize reliably to novel category-level objects whose shapes differ significantly from the training object. Third, the digital-twin tracking and generation add some computational overhead that may limit real-time deployment on resource-constrained robots.

VI. CONCLUSION

In this work, we address the challenge of viewpoint generalization in single-view 3D visuomotor policy learning by introducing real-time digital-twin augmentation for point-cloud observations. Our approach substantially reduces the reliance on additional multi-view data collection by dynamically simulating complete observations based on tracked object poses and a robot model. Through extensive simulation and real-world experiments, we demonstrated that this augmentation strategy significantly improves policy robustness under novel viewpoints. Furthermore, our analysis on the spatial relevance of different robotic arm components provides practical guidance for selecting informative components in manipulation observations. Future work could focus on improving digital twin prediction under object occlusion, searching for model-free object tracking approaches, and enhancing shape generalization, allowing the use of more generic object models.

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